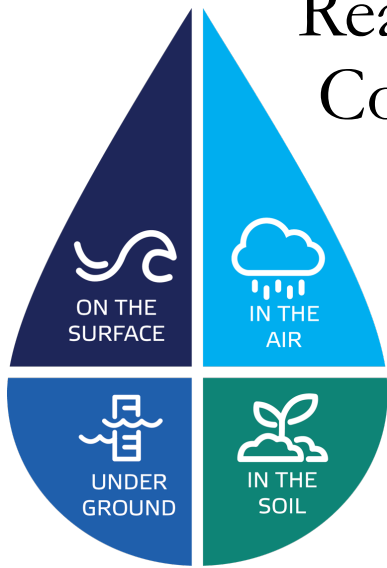


Soil Water Retention Curve

Real Time Measurement of Unsaturated Hydraulic Conductivity Using In Situ Dielectric Permittivity



Soil water movement
Laboratory to the Field

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Soil Water Movement Has Major Influences

- Irrigation Management and Optimization
- Phytoremediation
- Low Tillage and Carbon Sequestration
- **Satellite Ground Truthing**
- **Drought Forecasting/Water supply forecasting**
- Dust Control
- **Land Slide Studies**
- Biofuels and Ag Research
- **Forest fire potential**
- Infiltration rates
- **Flood Warning**
- Restoration
- **Climate Change Studies**



The Unsaturated Hydraulic Conductivity

The Kunsat, is an important parameter for modeling and predicting soil moistures and soil water movement.

The goal of this work is to calculate the Kunsat in real time from a meteorologic station equipped with a Stevens HydraProbe Soil Sensor



Soil Water Movement

Water moves through soil four different ways

Flow	Governing Equation	Application
Saturated Flow	Darcy's Law	Aquifers and saturated soil
Steady State Flow (unsaturated flow where wfv doesn't change with time)	Buckingham-Darcy Equation	Movement of water in and out of the capillary fringe. Under pavement where ET is small.
Transient Water Flow	Richards Equation	Agriculture, ET Driven Flow
Vapor Phase water movement	Fick's 2 nd Law	Dry conditions below soil water residual. Water moves from high to low concentrations as a function of time. Driven by the Gibb's Free Energy

One Dimensional Richards Equation

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(K \left(\frac{\partial h}{\partial z} \right) \right) + \frac{\partial K}{\partial z} - S \quad [1]$$

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(K \left(\frac{\partial h}{\partial z} \right) \right) - ET \quad [2]$$

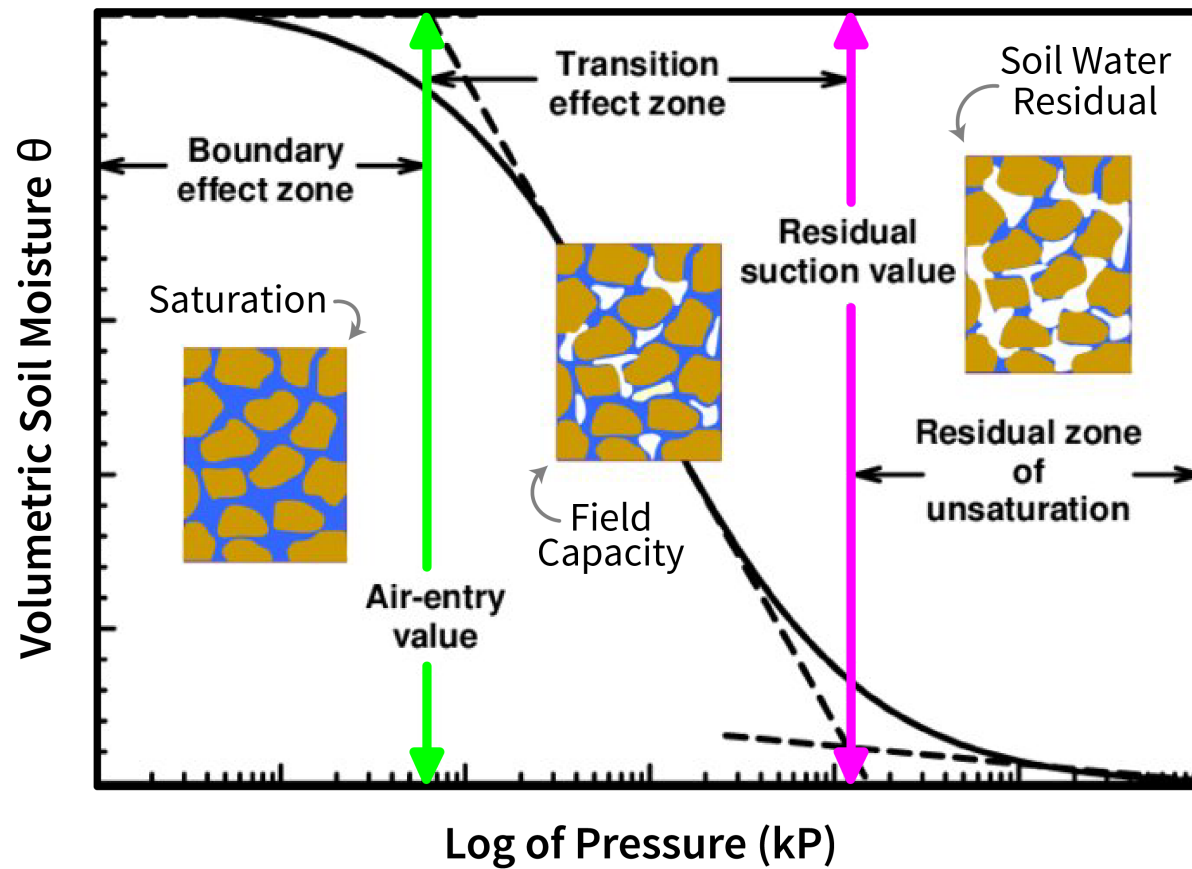
$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad [3]$$

$$K_{\text{unsat}} = K_s S_e^\ell \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2 \quad [4]$$

$$\frac{\partial}{\partial z} \left(K \left(\frac{\partial h}{\partial z} \right) \right) = W^{(t2)} = K_{1+1/2}^{(t2)} \left(\frac{h_1^{(t2)} - h_2^{(t2)}}{\Delta z^2} \right) - K_{2+1/2}^{(t1)} \left(\frac{h_2^{(t2)} - h_3^{(t2)}}{\Delta z^2} \right) + \frac{K_{1+1/2}^{t2} - K_{2+1/2}^{(t2)}}{\Delta z} \quad [7]$$

- Richards equation describes transient water movement.
- It can be numerically solved with a soil water retention curve, Ks or Ksat and time series metrological data.
- With numerical methods (first published by Celia 1990).

Soil Water Retention Curve Soil Moisture Vs. Hydraulic Head



van Genuchten Equation

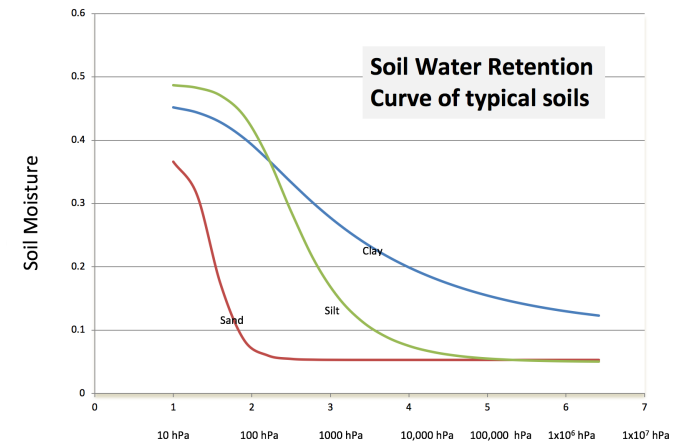
Water Content Modeling and Prediction

$$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{[1 + (\alpha|h|)^n]^m}$$

$$S_e(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r}$$

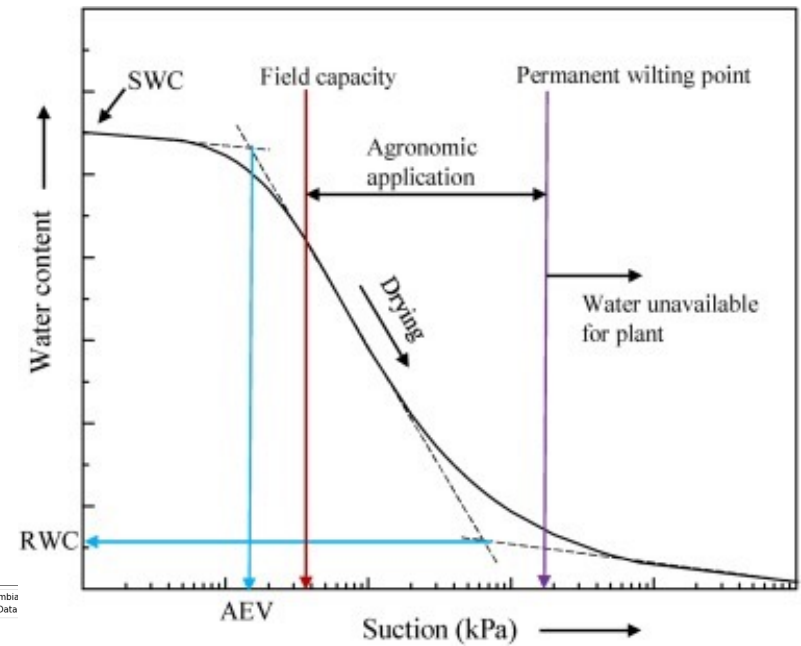
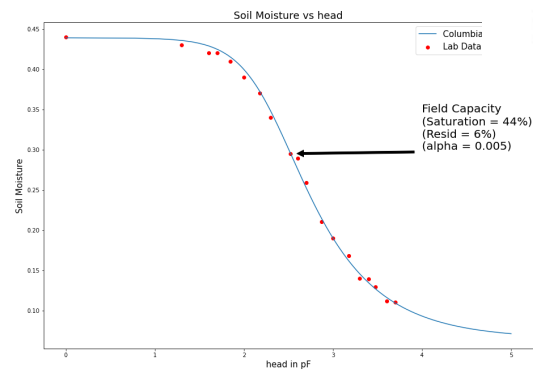
$$K(h) = K_s S_e^{\ell} \left[1 - \left(1 - S_e^{1/m} \right)^2 \right]$$

$K(h)$ or K_{unsat} = The Unsaturated Hydraulic Conductivity
[a parameter that indicates how well water moves through soil]



Symbol	Definition
θ_r	Soil water residual. A low level of soil moisture where the microscopic channels of water in soil become disconnected.
θ_s	Soil moisture at saturation.
α	Air entry value. Point below saturation where the air in soil becomes displaced.
n	Fitting parameter that is soil specific. ($m=1-1/n$)
h	Hydraulic head which is basically the same thing as the matric potential.

Water Retention Curve Applied to Real Time Data from the Field

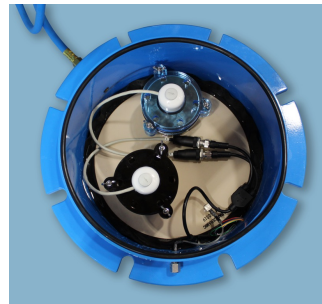
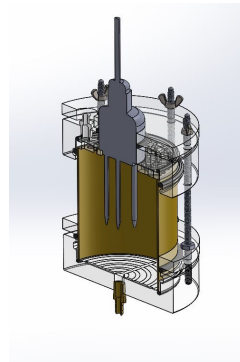
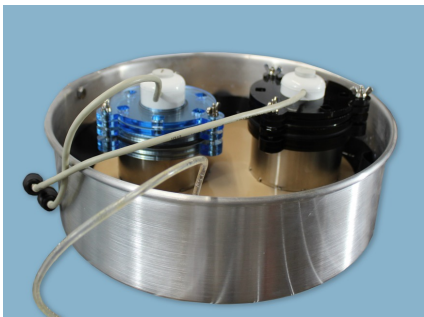


Determine a Water Retention Curve While Logging Data with a HydraProbe, The Travel Assembly



Portable Soil Sample that can be moved to many data collection stations.

- Tempe Cell
- Low tension tables.
- the Balance
- Ceramic pressure plate extractor.
- Contains a HydraProbe Soil Sensor

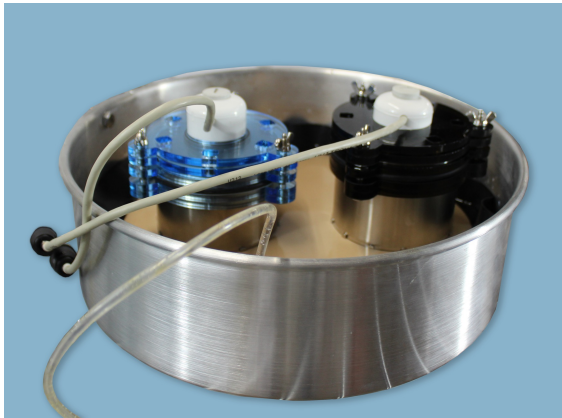


Low Tension

-0.01 to 1.5 Bar of negative Pressure from hanging water column

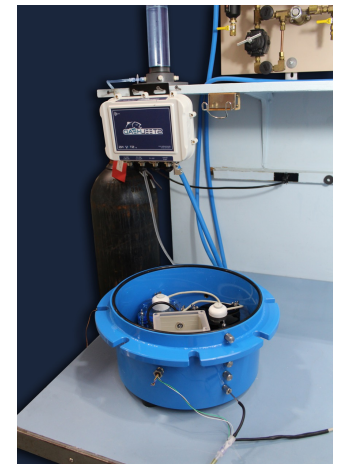
-Samples rest on porous partitions for good hydraulic continuity

-Adjustable reservoir to precisely control negative pressure.



High Pressure Extractors

- Positive Pressures up from 0.5 to 15 Bar (50 to 1500kPa)
- Samples sit on ceramic plate
- Pressure is regulated
- Lead through communication cable for data collection inside the extractor.



Test Soil

Wilsonville
Oregon

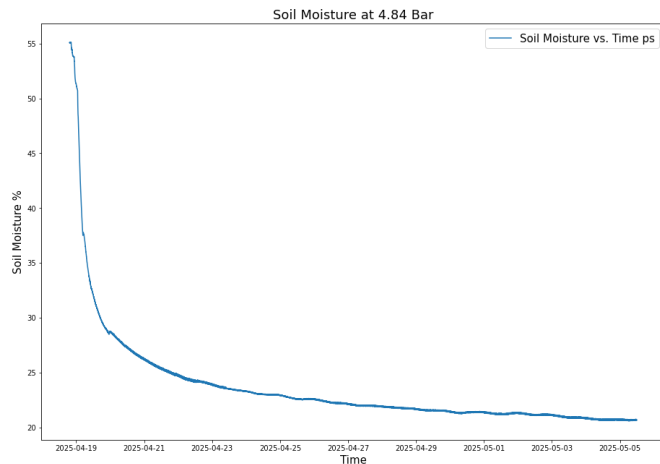


- Locations: State Oregon in the United States
- Crops: typically blueberries and grapes
- Soil is a well drained silt loam Columbia Silt Loam Alfisol ice age flood sediment



Advantage of having real time soil moisture measurements during pressurization

- Can reduce labor and run time
- Eliminate uncertainty about completion time
- Monitor the progress to detect problems



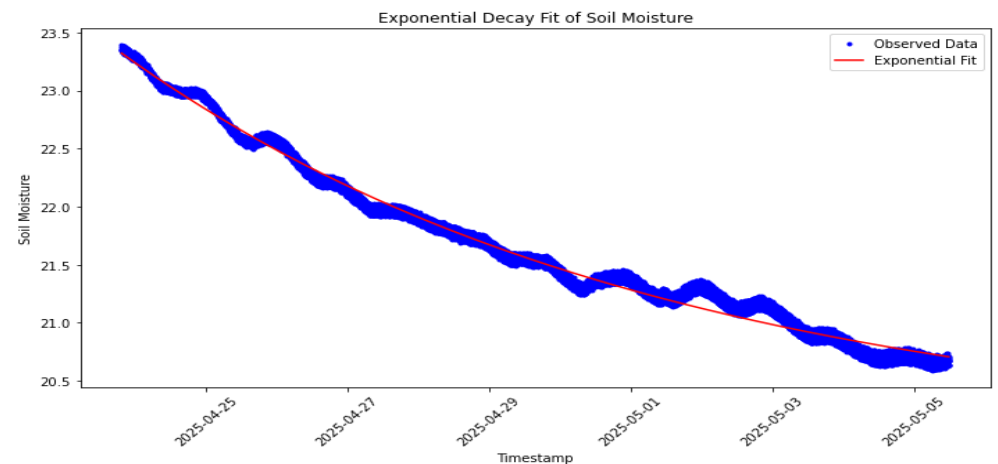
Exponential Decay of Soil Moisture

$$A = 3.3279$$

$$k = 0.0000002$$

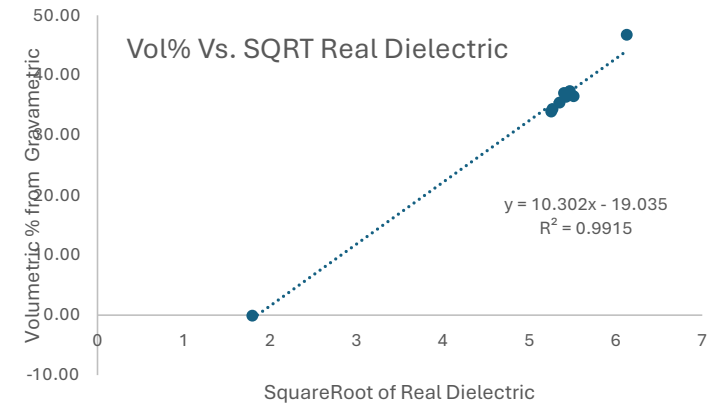
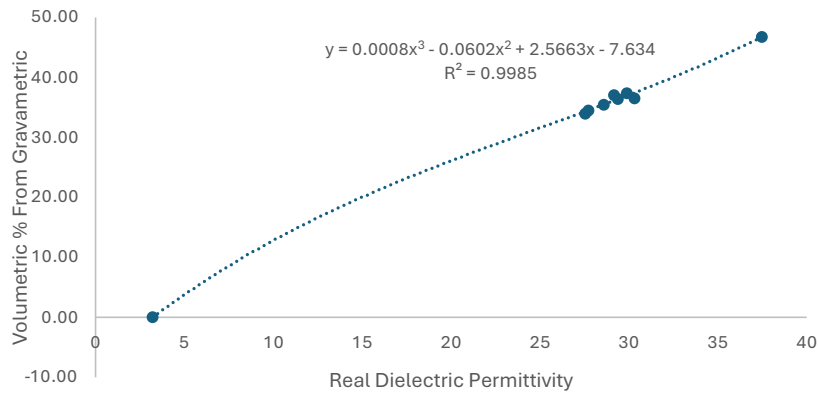
$$C = 19.9941$$

The R^2 between soil moisture and the curve fit decay of soil moisture every 6 hours is 0.992



Custom Calibration

Vol% Vs. Real Dielectric, Third order Polynomial



Custom Calibration N=9, Three Trials Each

Method	Formula	R ²	RMSE
Factory (Seyfried Equation)	$\theta = A + B\varepsilon_r^{0.5}$	0.9863	+/-1.44%
Third Order Polynomial	$\theta = A + B\varepsilon_r + C\varepsilon_r^2 + D\varepsilon_r^3$	0.9985	+/- 0.16%
Semi-Linear, Square root of real Dielectric	$\theta = A + B\varepsilon_r^{0.5}$	0.9915	+/- 0.38%

Custom Calibration can be applied to the van Genuchten equation with an accuracy of $\theta > 1\%$ and 0.016%

$$h = \frac{\left(\left[\frac{\theta_s - \theta_r}{\theta - \theta_r} \right]^{1/m} - 1 \right)^{1/n}}{\alpha}$$

$$K_{\text{unsat}} = K_s S_e^{\ell} \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2$$

Probe Accuracy	+/-1% Head	+/-0.16% Head	+/-0.16% K Unsat
Head Accuracy at 40% Moisture	+/-18 cm	+/-3 cm	+/- 4%
Head Accuracy at 30% Moisture (300 cm)	+/-30 cm	+/- 5 cm (+/-1.6%	4.4%
Head Accuracy at 20% Moisture (1000 cm)	+/-100 cm	+/- 17 cm (+/1.7%	7%
Head Accuracy at 15% Moisture (1800 cm)	+/-300 cm	+/- 50 cm (+/-2.8%)	10%

Conclusion

- Understanding and modeling the movement of water through soil has many applications from emergency response to land management.
- A Soil Water retention curve is critical for understanding the movement of water through soil
- A device called a Travel Assembly was developed to associate the water retention curve to an in situ soil moisture sensor for real time analyses.
- Determination of a water retention curve while a soil sensor in the sample provided an accuracy of $\pm 0.16\%$ with an R^2 of 0.9985 and under 10% for Kunsat up to 15 bar.

Thank You

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